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Self indexing mule shoe mechanism

Abstract

A system and method for providing a rotating a mule shoe assembly. The mule shoe assembly is disclosed with a top sub adapted to be connected to a tool string. A spring retainer for retaining a wave spring in a chamber created by the top sub and spring retainer. A mule shoe adjacent the spring retainer that linearly translate and rotate when a force is applied. The spring retainer has one or more pins that engages one or more linear slots in the top sub. A key moves in an indexing slot causing the mule shoe to linearly translate and rotate. The indexing slot is machined in the outer diameter of the top sub. During the stroke of the mule shoe, the one or more pins move in the linear slots and the key moves in the indexing slot.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present document is based on and claims priority to India Provisional Application Serial No.: 202311028771, filed Apr. 20, 2023, which is incorporated herein by reference in its entirety.

BACKGROUND

(2) This invention relates to mule shoe mechanisms in generally well operations. More particularly, the invention relates to a method and apparatus for using a rotating mule shoe.

(3) Downhole tools often need a mule shoe at the lower end of a tool string. The mule shoe is typically connected to the leading end of the tool string to guide the tool string in a wellbore. The mule shoe has a nose which is shaped, typically tapered, to push aside deposits and debris and fluids as it is run down hole into the wellbore. Prior art mule shoes may have a standard compression helical spring. In order to prevent the mule shoe from being stuck or provide proper orientation, the tool string would have to be rotated. The rotation of the tool string would cause the standard compression spring to distort and twist compromising the effectiveness of the spring. The prior art does not have a mechanism which prevents the rotation of a spring due to twisting moment.

(4) What is needed is a way to prevent the rotation of the spring due to twisting. The current invention uses a wave spring and a rotatable mule shoe in the mule shoe assembly to solve this issue.

SUMMARY

(5) The current invention is a self indexing mule shoe assembly. The mule shoe assembly is adapted for connection to an end of a tool string comprising a top sub adapted for connection to the end of the tool string. A spring retainer for retaining a wave spring in a chamber created by the top sub and spring retainer. A mule shoe adjacent the spring retainer that linearly translate and rotate when a force is applied. The spring retainer has one or more pins that engages one or more linear slots in the top sub that allows the spring retainer to linearly move. The top sub has a first end with a threaded section adapted to connect to end of a tool string and a second end of the top sub is connected to the mule shoe, wherein the second end is not threaded. The mule shoe linearly translates and rotates by a key on the mule shoe that engages with an indexing slot. The indexing slot and the key allows for a full 360 degree rotation. The key connects the top sub to the mule shoe. The wave spring is positioned between an upper stop created by the top sub and a lower stop created by the spring retainer.
(6) A method of deploying a tubular in a wellbore comprising connecting a mule shoe assembly to an end of a tool string. Running the tool string and mule shoe assembly into the well. Actuating the mule shoe assembly by stroking a mule shoe by rotating and linearly moving a mule shoe relative to a top sub when a compression force or tension force is applied. Rotating and linearly moving the mule shoe comprises applying a compression force or tension force to the mule shoe to move a key on the mule shoe along a path created by an indexing slot machined on the top sub. The compression force created is by an obstruction in the wellbore or the tension force is created by a wave spring, the compression force created by the obstruction compresses the wave spring and the tension force expands the wave spring. Each stroke of the mule shoe causes the mule shoe to move linearly and rotate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Certain embodiments of the disclosure will hereafter be described with reference to the accompanying drawings, wherein like reference numerals denote like elements. It should be understood, however, that the accompanying figures illustrate the various implementations described herein and are not meant to limit the scope of various described technologies. The figures are not necessarily to scale, and certain features and certain view of the figures may be shown exaggerated in scale or in schematic in the interest of clarity and conciseness.
(2) FIG. **1** is a cross-sectional diagram of an indexing mule shoe assembly in an uncompressed position.
(3) FIG. **2** is a cross-sectional diagram of indexing mule shoe assembly in a compressed position.
(4) FIG. **3** is an orthogonal side view of a top sub of the indexing mule shoe assembly illustrating an indexing slot and linear slot.
(5) FIG. **4** is a cross-sectional diagram of the top sup, spring, spring retainer and screw of the indexing mule shoe assembly.
DETAILED DESCRIPTION
(6) The invention will now be described in detail with reference to a few preferred embodiments, as illustrated in the accompanying drawings. In describing the preferred embodiments, numerous specific details are set forth in order to provide a thorough understanding of the invention. However, it will be apparent to one skilled in the art that the invention may be practiced without some or all of these specific details. In other instances, well-known features and/or process steps have not been described in detail so as not to unnecessarily obscure the invention. In addition, like or identical reference numerals are used to identify common or similar elements.
(7) FIG. **1** illustrates an indexing mule shoe assembly **10** in which the apparatus and method of the invention may be employed. The mule shoe assembly **10** includes an elongated top sub **12** and a

spring retainer **16** located external and adjacent to the top sub **12**. A chamber is created between the top sub **12** and spring retainer **16**. Within the chamber is a wave spring **14**. The wave spring **14** may act as a shock absorber as the mule shoe assembly **10** descends in a wellbore. Additionally, the wave spring **14** assists with orientation of the mule shoe **22**. The wave spring **14** allows the mule shoe **22** to rotate and translate linearly. The mule shoe assembly may be used with a completions string to ensure an unrestricted and smooth guided entry into a polished bore receptacle or lateral wellbore. Additionally, there are no internal threaded connection between the top sub **12**, spring retainer **16** and mule shoe **22**.

(8) The top sub **12** has a first end **38** and second end **40** as illustrated in FIG. **3**. The top sub **12** is generally cylindrically-shaped with a bore. The top sub **12** is tubular shaped and has three section with three different circular cross-section diameters. The three different circular cross-section diameters of top sub **12** creates two shoulders. A first section **32** incorporates the first end **38**. The first section **32** is the largest of the three sections. The second section **34** is between the first section **32** and third section **36** of the top sub **12**. The first end **38** has a thread section (not shown) which is adapted to connect to a threaded lower end of a tool string or another tool. The first end **38** of the top sub **12** may be connected to a downhole tool above the top sub **12** such as a completion or measuring tool. The second end is not threaded and is adapted to be connected to the mule shoe **22**.

(9) Connected to the lower end of the top sub is the mule shoe **22**. The mule shoe **22** is rotatable. The mule shoe has a nose mounted at the lower end of the mule shoe. The mule shoe **22** may have a tapered nose section. The taper extends from a wall of the mule shoe **22** to lower end of the mule shoe **22**. The nose is adapted to facilitate entry into a tubular in a lateral wellbore or a polished bore receptacle. Additionally, the tapered nose section is adapted to rotate when the tapered nose section is not aligned for entry into a lateral wellbore or a polished bore receptacle. The tapered nose section is adapted to breakup debris.

(10) FIG. **2** illustrates the indexing mule shoe assembly **10** in a compressed position. The mule shoe **22** is compressed relative to the top sub **12**. The mule shoe acts as a piston causing spring retainer **16** to travel towards the top end **38** of the top sub **12** compressing the wave spring **14**. In the compressed position the mule shoe **22** has traveled linearly and has indexed causing the mule shoe **22** to rotate. Furthermore, the spring retainer **16** also travels towards the top end **38** of the top sub **12** compressing the wave spring **14**. The mule shoe **22** can be compressed and uncompressed many times cycling the mule shoe **22**. The wave spring **14** assisted with uncompressing the mule shoe **22** and rotating the mule shoe **22**. The tool string does not have to be rotated to orient the mule shoe **22** since the indexing mule shoe assembly **10** is able to rotate without the assistance of the tool string.

(11) FIG. **3** illustrates a linear slot **24** in the third section **36** of the top sub **12**. The spring retainer **16** has a pin **18** that engages the linear slot **24**. There may be more than one linear slot **24** in the third section **36** and corresponding pins **18** in the spring retainer. The linear slot **24** and pin **18** guide the spring retainer **16** as the spring retainer moves linearly between a first and second position. The one or more pins **18** eliminate the twisting moment being transferred on to the wave spring **14**, thus preventing the damage of the wave spring **14**. The linear slot **24** determines the distance the spring retainer **16** travels between an uncompressed and compressed position of the mule shoe assembly **10**. The top sub **12** has an indexing slot **26** cut in the outer surface of the top sub **12**. The indexing slot **26** is a guide for the rotation of the mule shoe **22**. A key **20** moves in the indexing slot **26** causing the mule shoe **22** to rotate. The key **20** connects the top sub to the mule shoe. The key **20** prevents the mule shoe **22** from falling into the wellbore. The indexing slot **26** is machined in the outer diameter of the top sub **12** so that the mule shoe **22** rotates as the mule shoe **22** moves linearly. The mule shoe **22** only rotates when the mule shoe assembly **10** is stroked switches from an uncompressed position to a compressed position and vice versa.

(12) FIG. **4** illustrates the top sub **12**, wave spring **14**, spring retainer **16** and the key **20**. The wave spring is positioned between an upper stop **30** created by the top sub **12** and a lower stop **28** created

by the spring retainer **16**. The spring retainer **16** travels linearly towards the top section **32** of the top sub **12** compressing the wave spring **14**. The pin **18** prevents the spring retainer **16** from falling into the wellbore.

(13) In operation, the mule shoe **22** makes the first contact with an obstruction in a wellbore causing the mule shoe **22** to translate and compressing the wave spring **14**. The compressive force generated rotates the mule shoe **22** and the key **20** is guided along indexing slot **26** machined on the top sub **12**. The indexing slot **26** is machined in the outer diameter of the top sub so that the mule shoe **22** moves linearly as the mule shoe **22** rotates.

(14) The one or more pins **18** engaged in the one or more linear slots **24** prevent the twisting moment of the mule shoe **22** being transferred to the wave spring **14** and spring retainer **16**, thus preventing the damage of the wave spring **14**. During the stroke of the mule shoe **22**, the pins **18** have a linear movement along the linear slots **24** machined on the top sub **12**.

(15) In the fully compressed position, the mule shoe **22** bottoms out on the top sub **12** thus transferring the compressive load from the mule shoe **22** to the wave spring **14**. Once the mule shoe assembly **10** has bypass the obstruction or removed the compressive force, the compression load is transferred back to the mule shoe **22** by the wave spring **14**. The angle of rotation in the current design is 42 degree however the angle of rotation can be adjusted. The appropriate angle of rotation ensures that the wave spring does not reaches the wave spring's solid length and eliminates the damage on the wave spring. The wave spring expands when the tool string is pulled up and the mule shoe rotates further by 48 degree. The current design of the assembly provides a full 360 degree rotation of the mule shoe.

(16) Although a few embodiments of the disclosure have been described in detail above, those of ordinary skill in the art will readily appreciate that many modifications are possible without materially departing from the teachings of this disclosure. Accordingly, such modifications are intended to be included within the scope of this disclosure as defined in the claims.

Claims

1. A mule shoe assembly adapted for connection to an end of a tool string comprising: a top sub adapted for connection to the end of the tool string; a spring retainer for retaining a wave spring in a chamber created by the top sub and the spring retainer; the spring retainer has one or more pins that engages one or more linear slots in the top sub that allows the spring retainer to linearly move; the wave spring is positioned between an upper stop created by the top sub and a lower stop created by the spring retainer; and a mule shoe adjacent the spring retainer that linearly translate and rotate when a force is applied by contacting an obstruction in a wellbore.
2. The mule shoe assembly of claim 1, wherein the top sub has a first end with a threaded section adapted to connect to end of a tool string and a second end of the top sub is connected to the mule shoe, wherein the second end is not threaded.
3. The mule shoe assembly of claim 1, wherein the mule shoe linearly translates and rotates by a key on the mule shoe that engages with an indexing slot.
4. The mule shoe assembly of claim 3, wherein the indexing slot and the key allows for a full 360 degree rotation.
5. The mule shoe assembly of claim 3, wherein the key connects the top sub to the mule shoe.
6. The mule shoe assembly of claim 1, wherein the wave spring is positioned between an upper stop created by the top sub and a lower stop created by the spring retainer.
7. The mule shoe assembly of claim 1, further comprising a downhole tool between the tool string and mule shoe assembly.
8. A method of deploying a tubular in a wellbore comprising: connecting a mule shoe assembly to an end of a tool string; running the tool string and mule shoe assembly into the well; the mule shoe assembly comprises: a top sub adapted for connection to the end of the tool string; a spring retainer

for retaining a wave spring in a chamber created by the top sub and the spring retainer; wherein the spring retainer has one or more pins that engages one or more linear slots in the top sub that allows the spring retainer to linearly move; a mule shoe adjacent the spring retainer that linearly translate and rotate when a force is applied; and actuating the mule shoe assembly by stroking the mule shoe by rotating and linearly moving the mule shoe relative to the top sub when a compression force is applied by an obstruction in the wellbore or tension force is applied by the wave spring once the obstruction is cleared.

9. The method of claim 8, wherein rotating and linearly moving the mule shoe comprises applying a compression force or tension force to the mule shoe to move a key on the mule shoe along a path created by an indexing slot machined on the top sub.

10. The method of claim 9, wherein the compression force created is by an obstruction in the wellbore or the tension force is created by a wave spring, the compression force created by the obstruction compresses the wave spring and the tension force expands the wave spring.

11. The method of claim 8, wherein each stroke of the mule shoe causes the mule shoe to move linearly and rotate.
